

Tutorial 11

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1 Ground State energy Shift of 1D SHO

Consider 1D SHO with the perturbation $H_p = g \cos(kx)$. Calculate the 1st order energy correction to the ground state $|0\rangle$ of the system.

2 Hydrogen atom under the influence of constant Electric field

A hydrogen atom in its ground state is in a constant uniform electric field that points in the z direction. The electric field polarises the atom. Show that there is no change in the ground-state energy of the atom to terms of first order in the electric field. The interaction energy is

$$H' = -\vec{p} \cdot \vec{E} = eEr \cos(\theta). \quad (1)$$

where $\vec{r}$ is a vector drawn from the nucleus to the electron.

3 Problem 3

Consider the Hamiltonian

$$\hat{H} = \begin{pmatrix} 1 + \lambda & 0 & 0 & 0 \\ 0 & 8 & 0 & 0 \\ 0 & 0 & 3 & -2\lambda \\ 0 & 0 & -2\lambda & 7 \end{pmatrix}. \quad (2)$$

1. Write the Hamiltonian in terms of $H = H_0 + H_p$.
2. Find first-order correction in energy.